

The Double Bind: Understanding Equity in the Context of Generative AI in Computing Education

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Abstract

This perspective paper critiques the lack of attention to equity in the growing discourse on integrating generative AI into the introductory computer science sequence (CS1). Drawing on both Warren’s interpretation of Afro-pessimism as an analytic lens and Gutiérrez’s four dimensions of equity: (1) access, (2) achievement, (3) identity, and (4) power, we explore the argument that generative AI is inevitable or “worth the trouble.” We argue that requiring the use of generative AI in CS1 rationalizes a clear double bind with material consequences for the equity of Black students. This perspective urges the computing education community to examine what is rationalized when technological progress is treated as self-evident.

CCS Concepts

• **Social and professional topics** → **Race and ethnicity**.

ACM Reference Format:

Ashlyn Campbell and Angela D. R. Smith. 2026. The Double Bind: Understanding Equity in the Context of Generative AI in Computing Education. In *Conference on Research on Equity and Sustained Participation in Engineering, Computing, and Technology (RESPECT 2026)*, June 08–10, 2026, Chicago, IL, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3796496.3811755>

1 Introduction

Since the 2022 release of ChatGPT, students’ adoption of generative artificial intelligence (AI) technologies has become a central focus in computing education [26, 41]. Many see this as an opportunity to personalize learning [3], while others raise concerns about potential harms [5]. Amid ongoing discussions and early findings, educators have begun integrating generative AI into foundational learning spaces such as introductory computing courses (CS1) [52]. In turn, there is a growing conversation about whether the next logical step is to reorganize CS1 entirely by positioning generative AI as the entry point to programming [45].

This refers to the “prompts first” approach presented by Reeves et al., which argues that the rise of generative AI represents the inevitable next step in the historical progression of computing toward higher levels of abstraction [45]. These levels of abstraction range from theoretical computing in the 1930s to object-oriented

programming languages (e.g., Python, Java, C#) introduced in the 1960s, and now to the emergence of *natural language programming*. Natural language programming refers to teaching CS1 students to use generative AI to create projects first, then learning the underlying code later. The authors claim this approach can remove syntax barriers, enable students to program in their own words, and broaden participation like never before [45].

While we do not dispute the claim that computing has historically moved toward higher levels of abstraction, we argue that positioning generative AI as a solution to these problems justifies a lack of consideration for equity [28]. Specifically, the “prompts first” approach overlooks that participation in computing is structured by a double bind: the very systems students learn within are inseparable from the conditions that produce the barriers they are meant to address [56]. Ultimately, this reflects a techno-solutionist logic, in which complex social, historical, and structural problems are recast as solvable through technical mediation alone [10].

Our perspective draws on both Calvin Warren’s interpretation of Afro-pessimism as an analytic lens and Rochelle Gutiérrez’s four dimensions of equity framework to critique the framing of generative AI as a vehicle or solution to long-standing problems in computing education [21, 56]. We use Afro-pessimism to express this double bind as the structural impossibility for Black people to obtain equity within existing systems, since attempted solutions are constrained by those same systems [56]. We also draw on Gutiérrez’s framework, which shifts analysis away from deficit narratives in STEM education toward students of color by bridging equity scholarship’s dominant axis (access and achievement) with a critical axis (identity and power) [19]. It is with intention that Afro-pessimism (a non-reformist critique) and Gutiérrez’s equity framework (a reformist and non-reformist critique) are used in tension to examine how equity claims break down for students and lead to justification in the pursuits of innovation.

Furthermore, we situate this perspective within the discourse of computing education to move the term equity beyond an “umbrella term” for access and diversity [28], toward a foundation that more fully represents the students we serve. In doing so, we center the experiences of Black students in computing, given the long history of anti-Black technologies and cultures in computing [36, 43, 44].

2 Positionality Statement

Recognizing that research is not conducted in a vacuum, we offer this positionality statement to acknowledge how our identities and standpoint shape our role in this work [22]. The first author is a Black woman and the first in her family to pursue an undergraduate and graduate education in computing. As an alumnus of



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ACM ISBN 979-8-4007-2510-4/2026/06

<https://doi.org/10.1145/3796496.3811755>

a Minority-Serving Institution (MSI) in the South of the United States, my experience informs my tenacity for equity in computer science. Nonetheless, I am aware of my responsibility at my existing prestigious research institution to bring light to the experiences of students who are often most impacted by solutionist approaches. In this paper, I draw on my personal experiences of being stuck in a double bind to examine each dimension of equity, highlighting parallels between everyday instances of inequity and those perpetuated in computing. I offer this perspective, as my identity provides an insider look for interrogating how natural language programming may harm Black students.

The second author is a Black woman scholar and Assistant Professor whose research racializes technology through cultural computing and race-based methodologies. Having spent over a decade examining how technologies designed for “everyone” routinely fail Black communities, I approach this work with both scholarly rigor and personal stakes. As an educator who teaches future information professionals, I witness how solutionist framings shape what students believe is possible, necessary, and inevitable—and this paper emerges from mentorship conversations in which we grappled with what it means to think critically about generative AI while institutions rush toward integration. I write not as a detached observer but as someone implicated in the systems we critique, accountable to Black students who must navigate whatever freedoms we collectively choose to institutionalize.

3 Why isn’t Equity Realized?

3.1 The Framing of Equity

We begin with a definition of *equity*, as it is often mistaken for equality, leading to various orientations and outcomes [27]. This distinction is often explained by noting that equality requires treating students equally, while equity requires providing the resources every student needs to succeed. Ultimately, this means that the same resource may be insufficient to ensure equal outcomes for everyone. However, researchers have argued that such explanations often assume a single form of redistribution, simplicity, and benign, fixed conditions [27]. Take, for instance, a student in an introductory computer science course who is struggling to complete programming assignments. A proposed solution may be an additional resource, such as generative AI, but in reality, the student may require different forms of support, such as a curriculum that engages their interests [55]. Yet, focusing only on adding resources rather than questioning or restructuring the learning environment itself tends to stem from a fixed imagination in achieving equity [27]. Therefore, the most correct definition of equity emphasizes fairness over sameness.

Gutiérrez’s equity framework across the dominant and critical axis elaborates on this definition [20, 21]. As part of the dominant axis, Access, which is the most popular dimension of equity, focuses on how students can “play the game” or thrive in the existing system. However, Gutiérrez argues that it is insufficient without the other dimensions because it assumes sameness [20]. While Achievement, another dominant axis, evaluates how successfully students “play the game,” Gutiérrez argues that this dimension is often used to place students of color in a deficit-based narrative labeled the “achievement gap,” rendering it just as ineffective [19].

As part of the critical axis, Identity attends to students’ capacity to “change the game”; for example, when students see their cultural experiences reflected in computing or are encouraged to bring their full identities into their work, they are less likely to feel pressure to conform to dominant norms [55]. Issues in the Identity dimension have revealed why merely participating in computer science (Access) and continuing in the ‘pipeline’ (Achievement) fail [1]. Gutiérrez argues that students must have a balance of self (mirrors) and others (windows) in the curriculum to truly consider this dimension [20]. Also within the critical axis, Power focuses on who has the authority to “change the game”; for instance, students may be invited to question narratives in computing, rather than simply adapting to them [25]. Power is a dimension that forces us to rethink the possibility of achieving equity without looking at how computer science as a field can change. Each of these dimensions is interchangeable; without considering all of them, the chances for equity break down.

3.2 The Double Bind

Calvin Warren’s interpretation of Afro-pessimism is a way of understanding anti-Blackness that focuses on what it means to exist at all [56]. It is important to first understand that Blackness is not a fixed or singular category. Instead, it can be understood as lived experience, as identity, and as a site of resistance and self-definition [6, 11]. To say “I am Black,” in this sense, is affirmation and resistance [9, 12]. However, Warren argues that within dominant philosophical traditions, Blackness is also positioned as “non-being” or “nothing,” and that modern conceptions of the human depend on this exclusion in order to define personhood, value, and reality itself. From this perspective, the double bind describes a structural contradiction [56]. Blackness must be recognized by the same exclusionary systems leading to solutions failing to eradicate the outcomes of anti-Blackness, since solution-oriented thinking depends on it.

This is what creates conditions in education where even the most well-intentioned still find it challenging to address problems. But what Warren writes that situates his perspective most in the growth of generative AI in computing education is that “the success within the failure is precisely the exposure of this double bind.” Meaning that what appears to be failure can also reveal the structure that produces it, so we must first acknowledge that the problem is a double bind. This is especially important as Cunningham et al. argue that *techno-solutionism* emerges from a failure to recognize problems as double binds [10]. Techno-solutionism refers to the belief that technological innovations alone can resolve complex societal challenges [35]. A solution requires making decisions about who counts and what harms or consequences are acceptable to researchers or educators [10]. In doing so, the creators of solutions implicitly define which risks are within scope and which are excluded from consideration. To understand who is the rationale in this case, we build on this logic for our argument.

4 What happens when you are the rationale?

As generative AI is integrated into CS1, it is necessary to examine whether Black communities function as the rationale for innovation and what that positioning means for equity. In what follows, we examine each of Gutiérrez’s dimensions not to propose how equity

might be achieved, but to reveal how the language of equity (across dimensions access, achievement, identity, and power) is mobilized in ways that reproduce the double bind [19, 21, 56].

4.1 Access & Achievement

In computer science (CS) education, *access and achievement* refers to the resources a student has at their disposal to engage with quality instruction and succeed [19]. Access is widely recognized in equity-focused research and is often denoted as prior programming experience [30]. Achievement is also a prominent dimension and is strongly associated with access [48]. This relationship between access and achievement has been shown to be a barrier to participation for underrepresented groups [31].

In terms of access, Reeves et al. briefly acknowledge potential challenges of natural language programming, such as paywalls that generative AI systems may introduce and over-reliance on novice students [45]. While this is concerning, as popular models such as ChatGPT have rapidly grown and now have subscription tiers with costs up to \$200 monthly, the justification occurs most directly in conversations of over-reliance. Specifically, it's been found that students with less programming experience before college are more likely to over-rely on generative AI as well as have a harder time adapting to generative AI in CS1 [32, 42, 52]. This pattern is all too familiar from other popular techno-solutionist approaches and referred to as the EdTech Matthew Effect, where affluent learners benefit the most from adoption [46]. In predominantly Black school districts, where there have historically been far fewer quality CS courses [30], this isn't just a challenge to be viewed as an "achievement gap," but one reflective of who has access to participate in pre-college programming courses.

In terms of achievement, these differences in access to CS before college among Black students translate into responsibility. Reeves et al. focus on these issues as they relate to computational thinking skills, which is a valid concern, but whose responsibility is it? The underlying rationale here is that the students are [16]. Recent work reveals that Black students are disproportionately surveilled and criminalized in their use of AI, while also being expected to adopt those same technologies for learning [50]. When generative AI is positioned as the first step or gateway to programming, Black students face a double bind: they must rely on tools that have historically been used against them while remaining fully responsible for any "misuse," because AI has no stakes. Even when that "misuse" or misconduct differs from department, to class, to instructor [2]. Flenady and Sparrow describe this as a growing "responsibility gap," where educators frame generative AI as a supportive tool or collaborative partner while placing full accountability for outcomes on learners [16]. Reeves et al. acknowledge limitations of generative AI as a tutor, such as hallucinations, pacing issues, and incorrect explanations, but do not address who bears responsibility for these failures [45]. As a result, learners are increasingly held accountable for their own success, while the promise of freedom of natural language programming remains uneven and contingent.

4.2 Identity

The third dimension of equity is *identity*, which refers to how students view themselves and how that sense of self is shaped

in educational environments [19]. In computer science education, this is often measured by computing identity or how students see computing as part of who they are [29]. Although computing does not equate to just programming, this is the primary marker of legitimacy in the field. This creates tension between students' sense of self, their values, and what is rewarded in the classroom, when they are also concerned about the impacts of technology [14, 53]. Vakil describes computing classrooms as spaces where political and ethical concerns are often pushed aside, and where students may feel pressure to separate their values from their academic and career goals [53]. This makes claims that generative AI is simply an inevitable entry point into CS1 more complicated, especially when the political nature of technology itself is not acknowledged [58].

Reeves et al. focus mainly on the learning challenges of using natural language programming with generative AI [45]. However, this focus leaves out the type of learning environment that such a shift creates and whether that environment treats technology as neutral or apolitical. Students entering CS1 often already have concerns about generative AI and its impacts, considering the hyper-visibility and history of AI and algorithmic bias in Black communities [4, 7, 36, 53]. For example, the environmental impact of training language models is often concentrated in Black communities [33]. McGee describes this as part of a longer history of environmental racism, where infrastructure such as data centers and landfills are disproportionately placed in predominantly Black communities [33]. In this sense, generative AI positioning generative AI as inevitable isn't a coincidence.

When these concerns are not acknowledged, students are placed in a difficult position. They can use the technology while setting aside their values, or resist its use and risk being seen as out of place in the classrooms where peer pressure to adopt is prevalent [38, 53]. From an Afro-pessimist perspective, this reflects an expectation that computing education should remain politically neutral, even with the entrance of socio-technical systems. When computing is presented as objective, students are taught to separate technical work from the systems that produce it [33]. This produces limited identity options, ranging from "I'm just an engineer" [15] to students who care about social issues but are not given meaningful ways to act on those concerns within **technical** computing spaces [53]. Vakil illustrates this through the experience of Lupe, a student who describes a social impact project as the first time she felt her interests fit within computing [53]. Before this, she felt she had to separate her values from her coursework, moving between spaces where she could express her identity and classrooms where she could not. Does consciousness about technology require us to live in two worlds? This highlights a limitation in Reeves et al.'s claim that generative AI simply enables students to use their own words, when in practice, it may also require students to leave parts of their identity outside of computing or force them to make a choice.

4.3 Power

The final dimension of equity is *power*, defined as students' "agency to affect change in school or society" [19]. Scholars have long argued that power must be central to equity-oriented analyses in education, particularly when examining socio-technical systems such as generative AI [54]. However, Reeves et al.'s argument largely

overlooks the power relations embedded in the risks they identify [45]. While risks are acknowledged, they are not examined in terms of who benefits, who is harmed, and most critically, who has a choice.

Choice becomes the most pressing issue in this context. In an era of rapid institutional AI integration in education, students are rarely positioned as agents who can meaningfully consent to or contest these changes. Especially given the added peer pressure to adopt generative AI in computing classrooms [38]. This reflects what Padiyath describes as the “illusion of inevitability,” a form of technological determinism that frames technological advancement as the primary driver of social change rather than human decision-making [37]. Within this framing, concerns about generative AI are often dismissed through claims that we must “embrace [generative AI] or be left behind” [37]. This rhetoric fails to acknowledge that students who are underrepresented in computing are disproportionately affected and alienated by these shifts [39]. In particular, Black students face heightened expectations of being “twice as good” to succeed in academic spaces while still being overlooked [57]. For these students, the stakes of generative AI adoption extend beyond learning outcomes to questions of survival in spaces that have historically questioned their presence and marked them as unintelligible [43, 44].

From an Afro-pessimist perspective, the rationale is that, for many Black students, AI adoption is not a choice but a condition for legitimacy within systems that narrowly define rigor [14]. As a result, when generative AI becomes the default starting point in a course, choice is effectively constrained. Therefore, student adoption of generative AI is no longer an option to be evaluated, but an obligation to be met and excelled at. In this sense, the reduction of student agency is not an unintended side effect of innovation, but a direct consequence of treating technological adoption as inevitable.

5 Why so Pessimistic?

One day in class, we had a long discussion about generative AI. Some of my peers were excited, talking through new ways these systems could be used. Others were hesitant. A question kept coming up: How can you rightfully use something that is tied to harm experienced by marginalized communities? I listened. Then someone said something that stopped me. A peer offered an analogy: We drive cars without knowing how they work. We don't need to understand AI either. It set off alarms in my head. I felt myself pull back. I didn't say anything. I didn't know how to explain, in that moment, that the analogy was insensitive. It assumed a relationship to mobility I had never been granted. So I sat and looked around for a look of validation from any of my peers. But the room was in agreement, and because I was afraid to be perceived as an angry Black woman, I stayed silent.

The comparison didn't describe freedom. It depended on it. It assumed participation was optional. That opting out carried no real consequences. That not knowing was harmless, rather than unevenly challenging. My peer didn't have to consider whether the car would become another site of scrutiny. Didn't have to calculate whether a single repair bill would mean choosing between gas and tuition. The analogy worked for them because freedom was their starting point, not their dream. As the conversation moved on, it became clear to me that this analogy relied on a kind of freedom I had never experienced.

For me, there was never a vehicle to “freedom.” My first car was the hardest thing I ever worked for. I was making \$7.25 an hour, and I saved just enough to buy it. Not because I wanted independence, but because that car and that job was how I was going to pay for school. Public transportation wasn't an option. My mother was busy.

And this was during COVID-19, when everything already felt closed off and uncertain. I didn't get a car because I wanted one. I got one because I had no other choice. And once I had it, the freedom everyone talks about never showed up. The car needed repairs almost immediately. It broke down often. I spent more money just to keep getting to work. I forced myself to want to drive. To enjoy it. To be comfortable behind the wheel. But I was scared every time. But I kept going. I worked. I saved. And most importantly, I put on a brave face. But the freedom was never there. The obligation was.

*I see the same promise generative AI presents. I see the same assumptions that freedom is equally the baseline rather than the aspiration. This is why we write from an Afro-pessimist position. To refute the demand for optimism at the expense of lived reality. Kihana Miraya Ross describes this kind of work as what is required in the “meantime, in between time”—that is, what can be done now to make the educational experiences of Black students more livable [18]. Afro-pessimism, in this sense, is not a rejection of possibility, but a refusal to accept demands of optimism at the expense of others. This vignette shares one lived experience of the first author and is not universal to all Black students, but an illustration of how “freedom” is differentially distributed. **When we ask whose suffering becomes the rationale for technological progress, we are asking whose experience of the car, the classroom, and the code is treated as the default.***

6 Implications

First, we are not arguing against generative AI. We argue against uncritically positioning generative AI as a solution to long-standing challenges in computer science (CS) education. Throughout this paper, we examined how equity is negotiated across access, achievement, identity, and power. While some claim these concerns are already past the point of being “worth the trouble” [45], Afro-pessimism compels us to ask whose suffering is normalized in the pursuit of a solution that is not structurally possible. As generative AI expands across K-16 education, we aim to motivate the CS education community to acknowledge that generative AI must be met with more than just optimism. To do so, we must 1) consider how our pedagogy limits student agency, 2) consider how our research community engages with equity-centered research, and 3) take the steps to shift away from techno-solutionism.

6.1 For Practice

First, implications for practice require moving beyond the idea that access and agency are synonymous. Educators must engage in critical reflection on their role in introducing generative AI into the classroom. Providing each student with access to generative AI does not constitute equity. Instead, we must draw on lessons from critical pedagogy to encourage agency in the classroom [17]. To do so, educators may consider how generative AI reshapes norms around rigor and competence, not just academic integrity. We must ask whether students are taught to question these systems or are simply expected to adopt them without relying on them. Educators

may consider how the systems they prioritize to show in class and discuss in the syllabus may be harmful. Specifically, are we defaulting to popular models such as ChatGPT, Gemini, and Claude, or are we also considering systems that engage explicitly with identity, such as ChatBlackGPT [13]? Educators might also consider whether students who choose not to use generative AI are perceived as less capable by peers and in the curriculum. In this light, ethics cannot be treated as an afterthought or neutral; it must be central in courses that incorporate generative AI [15, 49].

6.2 For Research

Second, implications for research demand greater attention to power within the CS education community [53]. As Nicki Washington stated¹, many scholars “were living and breathing the work of equity and inclusion long before it became ‘the new black’ for those who can conveniently step in and out of this work.” Yet Black and women scholars who have been disproportionately affected by recent cuts to diversity, equity, and inclusion (DEI) funding [34] are forced to continue to “speak or not speak” [51]. At the same time, research spaces that resist paradigms designed to support students underrepresented in computing continue to marginalize critical approaches [47]. When scholars who sustain equity-centered research are defunded and excluded, the consequences extend beyond the research. The downstream effect is that students lose the researchers and educators most likely to recognize [8, 40], “talk back,” and resist the dynamics described in this paper. If those committed to interrogating not just access and achievement, but identity and power are pushed out, then the structures that constrain Black students’ agency in computing remain unchallenged. To care for our Black students, we must care for the environments and role models that cultivate and support them [24].

6.3 For A Mindset Shift

Finally, this paper calls for a shift in mindset. Reeves et al. draw on a subset of the history of programming to suggest that natural language programming is the next logical step in an ongoing trajectory of abstraction [45]. That historical account is not incorrect, but it is only one history. It is a history that only gives space from the vantage point of abstraction that has always functioned as expansion rather than exclusion. It narrates progress without confronting how anti-Blackness has shaped who was permitted to participate in, define, and benefit from that progress. Afro-pessimism interrupts this technologically deterministic narrative of inevitability. It reminds us that “anti-Blackness is no glitch” [23] in structures built for it. If that is the case, then no technological intervention can function as a complete solution. At best, it can be a step, provided the underlying double bind is acknowledged. As Tiera Tanksley urges [50], we must ask: Critical of what? Ethical for whom? And most importantly, efficient to what ends?

6.4 Concluding Thoughts for the RESPECT community

In the pursuit of a solution, a problem’s complexity is often simplified. However, we urge the RESPECT community to challenge the

prevailing narrative that AI is inevitable until equity (across access, achievement, identity, and power) is as important as innovation. Our argument, while pessimistic, is not a dismissal of Reeves et al., but a call to examine the assumptions underlying their position. We offer this perspective to ensure that progress in computing education is measured not only by innovation, but by all dimensions of equity.

7 Acknowledgment

Thank you to the RESPECT community for continuing to address equity in computing and to Aadarsh Padiyath for his advice on this paper.

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